

9.0 What will you learn in this lesson

By the time you get to the end of this unit you should be able to :

1. Select a sub strand that needs audio support
2. Generate learning outcomes/ experiences for the sub strand
3. Prepare the teachers' notes
4. Prepare an audio script
5. Record the audio programme
6. Edit the audio programme
7. Suggest support materials for the audio programme
8. Operate a given audio equipment
9. Identify sources of learning resources for an audio programme

ASSIGNMENT

In groups of not more than 5 and a common subject

- (i) Identify a sub strand / sub topic that in your opinion needs audio support.
- (ii) After deciding on the programme level, identify the instructional outcomes
- (iii) Prepare the Teachers' Notes.
- (iv) Prepare a script, rehearse it and edit it to be between 15 and 20 minutes.
- (v) Develop support materials for the programme.
- (vi) Record the programme on a compact disc, memory card or flash disc
- (vii) Playback the recording and listen to it.
- (viii) Hand in the recorded material, the script, the Teachers' Notes and the support materials (support material is optional) for assessment.

9.1 Introduction

The process of producing audio recordings can be discussed under three major stages:

- (a). Planning
- (b). Script writing
- (c). Presentation / recording

9.2 Planning

The planning stage starts with the selection of strands /topics to make up the series. A series of topics for a particular class is made of topics enough to cover a year. In some countries a series is made up of 24 audio lesson topics namely eight per term. The selection is guided by the specific functions of the medium as well as the needs of the subject or course. We must bear in mind the weaknesses and strengths of the audio / television

medium as we select these topics. We must also give priority to topics that are not well covered in the class textbooks as these are the strands/ topics that actually need the audio support. The need arises due to scarcity of instructional materials and the nature of the content. Audio recordings can for example be useful in the teaching of music where the class teacher cannot demonstrate some of the tunes in the classroom or in the teaching of literature where different scholars give different interpretations of the same novel .throughout the selection the writer must consult with the subject specialist.

ACTIVITY 9.1

Recall the five senses you learnt in your earlier courses. Discuss the weaknesses of the audio medium as it applies to some of the senses. How would you make up for these weaknesses in the audio program you intend to produce.

ACTIVITY 9.2

For a named strand in one of your teaching subjects, identify at least two substrands that may benefit from audio support . Justify the choice of the substrands. Describe the nature of the support.

The next step under the planning stage is writing the programme outline for each sub strand/ subtopic. This outline is made up of six elements. The **first element** is the programme title which must be concise and precise . For example the titles “ Coffee growing “ and “Coffee growing in Kenya “ are different since in the former , the programme cannot confine itself to coffee growing in one country only or region but must address itself to coffee growing globally . The **next element** is to generate the specific learning outcomes (objectives) we wish to achieve through the programme . These will depend on the class level the programme is intended for. It is important in writing these learning outcomes to cater for the different key performance indicators (domains of educational objectives ie cognitive, psychomotor and affective domains). The outcomes must be achievable in the time available for the programme . The outcomes if well stated become very useful in content selection.

ACTIVITY 9.3

For one of the sub strands identified in activity 2, generate learning outcomes you expect to achieve by the end of the planned audio recorded material.

The **third element** of the programme outline is the content summary. Here the writer must now list the main facts, ideas , skills and attitudes to be covered by the programme . This content must be arranged in a logical manner for easy audio lesson development. The content summary if well prepared at this stage helps a great deal in deciding on the reference materials, resource persons to be interviewed and the illustrative examples if any. Once the content summary is in place it is then possible to decide on the **fourth element** of the outline namely form and structure. Deciding on

the form and structure involves choosing whether to use straight talk, illustrated talk , interview , panel discussion , documentary techniques , dramatization , or a mixture of the above (magazine) . The writer can then concentrate on the **fifth element** of the outline namely developing support materials to accompany the recording. The materials should encourage active participation of the learners in the programme. The material must involve learners at three critical stages of the lesson. The learning activities make up the **sixth element** of the outline. These stages are activities before the listening, during the listening and after the listening. Once the outline for the topics have been completed, it is advisable to discuss them with other subject specialists as to the validity and relevance of the proposed content .

The programme outline which comprises the sub strand, learning outcomes, content summary , support materials , and learning activities will be properly written out to make up the Teachers' notes. The Teachers' notes is the document that will guide the classroom teacher on how to make effective use of the audio recording.

SAMPLE TEACHER'S NOTES

CLASS: Standard 7
SUBJECT Home Science
TOPIC: Basic food groups for a balanced diet

Learning outcomes

By the end of the radio lesson, the learner should be able to:

- (i) Classify common foods into 3 food groups
- (ii) State functions of foods in each group.

Learning Aids

A chart showing drawings and mountings of the three food groups.

Activities before the broadcast

Pupils should be asked questions such as the following by the class teacher:

- (i) Name 3 good eating habits
- (ii) State at least 4 factors that influence eating habits
- (iii) Name 3 ways of obtaining food

The class teacher should also write the following on the chalkboard:-

Carbohydrates - energy giving foods

Proteins - body building foods

Vitamins - protective foods

Activities during the broadcast

The class teacher should receive responses from pupils when asked to do so by the radio teacher. S/he should quickly write down these answers. The teacher should display the chart showing drawings and mountings of the 3 food groups. S/he should point at the relevant parts when asked to do so by the radio teacher

Activities after the broadcast

The classroom teacher should ask pupils to write in their exercise books all the foods they eat at home and finally classify them. The pupils can also be asked to choose foods from these three groups and plan balanced meals for a whole week.

Once the script has been written and Teachers' Notes compiled, the script editor should edit the programme. The editor should consider the flow of content, the time slot allowed for the programme and the general layout of the script. He is to cross check the appropriateness of the recorded inserts and the music use. He also ensures that the signature tune chosen is appropriate. The signature tune is that piece of music or vocals that comes at the beginning and at the end of a programme. It is the same for a whole programme series and is regarded as the "trademark" of the programme series it identifies the programme. Usually the message in the signature tune is related to the theme of the series. The signature tune should never be used as bridge music. If the introductory remarks of a programme are to be voiced over the signature tune then the tune should be instrumental for it to be faded under the speech effectively.

ACTIVITY 9.4

Prepare the teachers' notes (manual) for the sub strand identified in activity 3.

9.3 Scriptwriting

The art of scriptwriting involves laying on paper a simulation of how people really speak. The key point to remember is that the audio channel is for the ear and not for the eye. The sentences must be short, consisting of familiar phrases since we rarely use long sentences in everyday speech. Long rumbling sentences must be broken down into shorter ones. The active quality of natural speech ensures that the scriptwriter captures the attention of the listener and conveys information. The best way to write an audio script would therefore be to put down word by word the way you would tell the story to a friend. The words must sound warm and personal to compensate for the loss of visual communication.

ACTIVITY 9.5

From one of the reference books you intend to use in teaching the identified sub strand, select a paragraph and rewrite it in the radio language.

The writer should consider whether the content suggests more than one presenter, a need for music, indigenous sounds, sound effects or simply one voice commentary. A good educational broadcast lesson should be voiced by a radio teacher and at least two radio pupils besides other presenters. It is good to have more than one presenter to give the programme some 'taste' but it comes but it becomes confusing when the voices are too many. Consideration should also be given to using sound effects at appropriate stages to give the programme a natural outlook for instance when the script is referring to an event taking place at dawn then the sounds of birds and other insects could bring in the time perspective.

If the script is referring to an interview involving a poultry farmer, then during the interview sounds of hens and cocks should be heard in the background. The programme must open and close with a signature tune. This is a sound effect that identifies the programme series, it should where possible show some relation to the content or theme of the programme.

The following rules should be followed in script writing:-

- (a). The script must convey the message in a logical manner. It should be made up of one's thoughts arranged in a sequence to drive the listener to stay tuned.
- (b). All the facts must be factual, i.e accurate, honest, current, truthful and sincere.
- (c). The script must be 'mobile', interesting and relevant to the environment. It must be illustrative so that the listener is mentally visualized.
- (d). The script writer must have a complete story in him.
- (e). A good script must have a 'hook' which is catchy, to catch the listeners attention and sustain it throughout the listening session.
- (f). The script must avoid overloading the programme with lots of data or dates that would confuse the listener.

The script layout should be standard as this helps in ensuring that all those involved in the production understand its form and structure. An audio script should be made up of four distinct parts. The first part should be the introduction. Here the radio teacher should welcome the listeners to the programme and connect it to previous knowledge. The second part is the main body of the programme where the content should be delivered logically, punctuated with the pupil activities. The programme should allow for pauses to give pupils time to respond to questions and tasks. The third part of the script is the closing. Here the radio teacher with the assistance of the radio pupils summarize the main points of the lesson. If necessary the next topic can be stated and the pupils asked to collect necessary materials. A series of questions to direct the follow up can also be posed. Finally, some programmes finish off by giving credit to participants namely the script writer, script editor, the presenters, the producer, the technician and the station.

It is important to remember that the script should be:

- (i). Typed, double or treble spaced with margins of 1 ½ to 2 inches for easy reading
- (ii). With pages numbered sequentially
- (iii). With each page starting on a new sentence
- (iv). Reproduced on single sided non-flimsy paper
- (v). With enough copies for all those involved in the production (producer , presenters , and technician)

Materials to be recorded in the studio should be separated from pre-recorded materials (taped inserts for interviews , music and sound effects). Identifying details should be given for all inserts including name of interviewee , title of music , description of sound effects , cue-in (opening words or sounds) and cue-out (closing words or sounds) and exact duration .

ACTIVITY 9.6

Discuss with your class mates

- The need to edit a script
- The need to choose an appropriate signature tune to the substrand or subject

- The need to choose an appropriate bridge music
- The importance of repeating learner's answers
- The importance of repeating questions , definitions
- The need to incorporate learning activities in the programme

SAMPLE SCRIPT

Programme series : Primary Home Science
 Programme title : Basic food groups for a balanced diet
 Programme level : Standard seven
 Programme number : 4 of term 1
 Script writers : Raphael Muruu, Selina Ominde, Eunice Obat, Gordon Oluoch
 Script-editor : Lordvicus W. Olanga
 Producer : Eunice Obat
 Technician : Jacob M. Oloo
 Duration : 14 minutes 25 seconds

STUDIO CAST

1. Radio teacher (.....name...)
2. Radio pupil 1 (.....name...)
3. Radio pupil 2 (....name..)
4. **SOUND EFFECTS:** Signature tune
: Bridge music

1. **CONTROL** :Bang on the signature tune...15 seconds.....then fade under
2. **ANNOUNCER:** We present Home Science for standard seven.This is the fourth programme this term.The title of today's programme is "Basic food groups for a balanced diet".D programme we will learn about :-
 (i). The definition of a balanced diet
 and (ii). The 3 basic food groups and their functions.

3. **CONTROL:** Fade up signature tune10 seconds...then fade out.
 4. **RADIO TEACHER :** Hello standard 7 pupils. In today's programme we are going to learn about the basic food groups that make up a balanced diet. You are required to give answers to our questions to your class teacher when asked to do so. With us in the studio are radio pupils Irene, Catherine and Sarah.
 5. **RADIO PUPILS:** Hello standard 7 pupils.
 6. **RADIO TEACHER:** We will start by asking you a question. What do you understand by the words 'balanced diet'? [Repeat]. Tell your teacher now.
 7. **CONTROL :** Fade up bridge music ...20 secs.....fade out
 8. **RADIO TEACHER:** Yes, Sarah.
 9. **SARAH:** A balanced diet is a diet which contains all the food nutrients required by the body, in our case carbohydrates, vitamins, proteins and fats.
 10. **RADIO TEACHER:** That is very good Sarah. Class, tell your teacher what you had for breakfast this morning. [Repeat]
 11. **CONTROL :** Fade up bridge music20 secs..... fade out.
 12. **RADIO TEACHER :** Yes, Irene.
 13. **IRENE :** I had bread and butter, a fried egg, a cup of tea and an orange.
 14. **RADIO TEACHER :** Good! Irene had bread and butter, a fried egg, a cup of tea and an orange. She had foods which give energy, build and protect the body. Irene had a balanced diet.
 15. **RADIO TEACHER :** Children, now tell your class teacher examples of foods that give us energy. [Repeat]
 - 16 **CONTROL :** Fade up bridge music25 secs..... fade out
 17. **RADIO TEACHER :** Now,Catherine tell us some examples of energy giving foods? [Repeat]
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18. **CATHERINE :** Energy giving foods are maize, cassava, potatoes and rice.
 19. **RADIO TEACHER :** Good! Catherine has told us that examples of energy giving foods are maize, cassava, potatoes and rice. Others are fats and oils from margarine, nuts and sugar. All these foods are carbohydrates. Now teacher.....display the chart showing the various food groups. (Repeat).
 20. **CONTROL :** Fade up bridge music..... 25 secs.....fade out
 21. **RADIO TEACHER :** Teacher,.....on your chart, show the pupils the group of foods that give us energy. [Repeat].
 22. **CONTROL :** Fade up bridge music.....25 secs.....fade out.
 23. **RADIO TEACHER :** The teacher has just shown you foods that give us energy . The second group of foods are the body-building foods. Class,..... Tell your teacher foods which build our bodies. [Repeat].
 24. **CONTROL :** Fade up bridge music.....25 secs.....fade out.
 25. **RADIO TEACHER :** Yes, Sarah, tell us some of the foods that build our bodies.
 26. **SARAH :** Body-building foods are milk, eggs, fish and meat.
 27. **RADIO TEACHER:** Good! Some examples of foods that build our bodies are milk, eggs, fish, meat, beans, peas and groundnuts. These foods are also protein-containing foods. They give proteins to our bodies. Now teacheron your chart show the pupils the group of foods that build our bodies. [Repeat].
 28. **CONTROL :** Fade up bridge music.....25 secs..... fade out
 29. **RADIO TEACHER :** Your class teacher has shown you foods that build our bodies. Now look at the chart again. (Pause 7"). You can see the third and last group of foods. What name is given to this group of foods. Tell your teacher.....now.
 30. **CONTROL :** Fade up bridge music.....20 secs fade out
 31. **RADIO TEACHER :** Yes, Irene.

32. **IRENE** : The third group of foods are called protective foods.
33. **RADIO TEACHER** : Protective foods provide our bodies with vitamins and mineral salts. Vitamins and mineral salts protect our bodies from certain diseases. Class.... Now tell your teacher some foods which protect our bodies.
34. **CONTROL** : Fade up bridge music....25 secs..... fade out
35. **RADIO TEACHER** : Yes,.....Catherine.
36. **CATHERINE** : Protective foods are carrots, tomatoes, vegetables, pawpaws and mangoes.
37. **RADIO TEACHER**: Protective foods protect our bodies from certain diseases and include carrots, tomatoes, vegetables, pawpaws and mangoes. Now before we end our radio lesson let us again look at the main points of the lesson . We learnt that foods can be classified into three main groups . What were these Tell your teacher now.
38. **CONTROL** : Fade up bridge music.....20 secs..... Fade out
39. **RADIO TEACHER** : Yes, Sarah.
40. **SARAH** : The three main groups are energy-giving foods, body-building foods and protective foods.
41. **RADIO TEACHER** : Good! The three main groups are carbohydrates or energy-giving foods,.....proteins or body-building foods,.....and protective foods which give us vitamins and mineral salts. These foods if eaten give us a balanced diet.Now,.....tell your teacher examples of energy giving foods . (Repeat).
42. **CONTROL** : Fade up bridge music..... 10 secs.....fade out
43. **RADIO TEACHER**: Yes, Irene.
44. **IRENE** : Energy-giving foods include maize, cassava, potatoes and rice.
45. **RADIO TEACHER** : Good! Now Tell your teacher examples of foods which build the body.
46. **CONTROL** : Fade up bridge music.....15 secs..... fade out
47. **RADIO TEACHER** : Yes, Catherine.
48. **CATHERINE** : Body-building foods include milk eggs, fish, meat, beans, peas and groundnuts.
49. **RADIO TEACHER** : Good! Class.....finally tell your teacher examples of protective or vitamin-giving foods . [Repeat].
50. **CONTROL** : Fade up bridge music.....15 secs..... fade out
51. **RADIO TEACHER** : Yes,.....Sarah.
52. **SARAH** : Protective foods include carrots, tomatoes, vegetables, pawpaws and mangoes.
53. **RADIO TEACHER** : From what we have learnt, it is important to include the three groups of foods in our daily meals. It is only then that we can say we have a balanced diet. A balanced diet ensures that we have enough energy, we build our bodies and protect our bodies from diseases.
54. **RADIO TEACHER** : In our next radio lesson, we shall learn about food preparation and serving. Between now and then your class teacher should arrange a visit to a nearby hotel to enable you to get an idea of how food is prepared and served. You can observe these at home . And that is all for today .
55. **CONTROL** : Bring up signature tune10 secs...then fade under.
56. **ANNOUNCER**: You have been listening to programme number 4 in the Primary Home Science series for standard 7. The title of the programme was "Food groups for a balanced diet." This programme was written by Raphael Murruu, Selina Ominde, Eunice Obat and Gordon Oluoch.It was edited by Lordvicius W. and produced by Eunice Obat. Your presenters were Muruu, Ominde, Oluoch and Ogolla, and behind the controls was Jacob Oloo. The programme was produced in conjunction with the department of Educational Communications and technology, Kenyatta University. Until next week, it's bye from us.
57. **CONTROL**: Fade up signature tune20 secsthen fade out.

The sample script shown above could be improved by say inserting an interview with a nutritionist or school matron, or including a conversation between a housewife and one of the presenters. As a means of enabling the classroom teacher to utilize the audio material effectively, the Teachers' Notes on the topic should be availed.

ACTIVITY 9.7

Prepare an audio script that would give a recording of 15 minutes.

9.4 RECORDING

Once the script has been written, edited and all the inserts and sound effects lined up then the next stage is to put the content on tape, that is recording. The producer must use voices that have been auditioned. During auditioning the voices are chosen for their quality in terms of pronunciation, clarity, fluency, pace, intonation and general voice quality. The voices should also be chosen for their suitability for narration, drama, poem reciting, and radio pupil roles.

Once the producer has assembled his artists (presenters) in the studio, each of them should have a copy of the script and thoroughly rehearse the script several times. This will help the producer and technician get an estimated duration of the programme. If there is need for further editing it can be done at this stage to reduce the programme to the prescribed duration and still have the message. During the recording session, the technician should ensure that the microphones are properly positioned to maintain a good recording level throughout the session.

If the duration of the recorded programme with all the inserts, music bridges and pauses included is still higher than the prescribed time, then post-production editing can be done. This can be accomplished through mechanical splicing or dubbing.

Editing

Editing a material means removing parts of it and re-organizing the material to give it 'shape'. Editing can be done in two ways:

Script Editing

Before recording a programme the writer should allow a colleague to read through the script and give comments. A media specialist should also be allowed to judge the script for its suitability. The writer can then modify the script in line with the comments given by the colleague and the media specialist. Sometimes the time slot allocated for the programme is 20 minutes and on rehearsal the programme overshoots the time. It is then necessary to cut down on the quantity of material to bring it down to under 20 minutes.

Tape Editing

This is an important aspect of recording. It is conveniently done in tapes than in cassettes. This is because of two reasons. Firstly, the tape speeds 7 ½ and 15 inches per second (ips) are ideal for editing than the slower audio cassette speed of 1 7/8 ips . Secondly the tape is one track while the cassette is two track .The following alterations can be done on tape programmes through editing:

- (i). Reducing the overall length of the programme
- (ii). Re-arranging the order of events by placing one sound sequence ahead or behind its original position on the tape , thus providing bases for comparisons , contrasts or emphasis .
- (iii). Inserting new material such as speeches , music etc .

There are two methods of tape editing :

Electronic Editing (Dubbing)

This is ideal for audio cassettes and involves copying (dubbing) one recording through a second recorder onto another tape or cassette. This is done without cutting the original tape / cassette. This method must be used very carefully as it introduces pops at the start and end of a sound sequence

Mechanical Editing (Splicing)

This method involves using scissors or tape slicer and splicing the tape. It is highly recommended for ‘polishing’ an interview by removing slight imperfections such as mmm eer , to add tape for lengthening pauses , reducing pauses or even to substitute a corrected bit of narration . The steps to follow are:

- (i). With the script in front of you listen to the recorded tape , listing spots that need editing and record the counter reading .
- (ii). Dub the tape onto another fresh tape, file the original tape for reference. Proceed with the tape you have just recorded.
- (iii). Lace the freshly recorded tape onto the recorder Play it and confirm the counter readings recorded earlier.
- (iv). Carry out rough editing first, that is removing whole paragraphs, whole sentences and rearranging them in the appropriate order. To identify a point to cut we engage the ‘edit’ key on the tape recorder, then we can move the spools manually back and forth across the play head.
- (v). The point to be cut can be carefully marked with a fine tipped felt pen or a china marking (grease) pencil.
- (vi). Two points on the tape can be connected using the splicing tape > the splicing tape should be put on the shiny side.
- (vii). After making a joint, playback the tape over the joint and satisfy yourself that it is well done.
- (viii). If a new bit is to be added, identify the spot where you wish to add the new bit . Mark it with a felt pen.
- (ix). Playback the tape over the two joints and satisfy yourself that the joints are well done and that the new bit fits in very well.

9.5 OPERATING AUDIO VISUAL EQUIPMENTS

Since the classroom teacher can decide to use instructional materials at any time during the teaching it is necessary for him / her to know how to operate the large variety of equipments . The teacher should not be over dependent on a technician as the institution may not be endowed enough to afford the needed technical staff. Since the institution may have various equipments, they should have a copy of a manual of operating instructions for each model of equipment, the manual contains details of equipment features and operating techniques . It is important that all users maintain a high standard of performance, this leads to the realization of the following benefits:

- Equipment can be kept to in working condition for a maximum amount of time
- The cost of equipment maintenance , repair and replacement can be minimized
- Incidents of the equipment failing to give the expected output can be reduced.

The sound produced should as much as possible resemble the original sound. The quality of the sound produced is affected by:

- The type and quality of equipment selected
- Operator’s skill in using the equipment.

Factors that affect the quality of sound reproduced during playback include:

- Characteristics of the original sound
- Environmental conditions in recording / playback
- Capability of the recording medium to capture all the frequencies of the original sound.
- Capability of the playback system to reproduce program material in full frequency and without distortion.

- Ability of the operator to manipulate the equipment
- Matching. All the components in an audio system should be of equal quality and have similar performance characteristics.

For operating most of the equipment, the teacher as well as the pupils must master the use of controls such as the normal play, record , fast forward (speed selection) , pause , rewind and stop buttons . In some cases the controls may be manual as well as through a remote control devise.

9.6 LOCATING SOURCES OF AUDIO VISUAL MATERIALS

Having discussed most of the audio visual resources, we have seen that some of them could be designed and produced by the teacher and / or pupils

, it is advisable to note that they are also available at the following locations :

- Kenya Institute of Curriculum Development (KICD)
- The British Council Libraries
- Teacher Advisory Centres
- Kenya National Library Services
- Learning Resource Centres of the local Teacher Training College.
- Community Media Centres (if there are any)
- Media Centres in other schools.
- Community resources available within the locality such as factories.

ACTIVITY 9.9

Visit one of the locations where instructional materials can be found. Enumerate the broad groups of audio visual materials that can be found at the location.